

College of Industrial Technology
King Mongkut's University of Technology North Bangkok

Seat No.

Midterm Examination of Semester 1

Subject: 392151 Chemistry I

Date: 2 September 2020

Year: 2020

Section: 15-18

Time: 15.00-17.00

Name: _____ ID: _____ Class: _____

Directions: The test is designed to measure your comprehension. The test is divided into 2 parts. There will be 10 pages (including this page) and they are worth 70 points.

- This exam paper contains no errors. If a suspected error is found, it is the student's discretion to correct it.
- Answer the questions on this test paper.
- Books, documents and lecture notes are not allowed.
- You must be in the room for one hour after the exam is started and, while taking the exam, you cannot go out except in an emergency case.
- Before leaving, make sure you do not bring this test outside.
- Do not use any electronic communication device.
- Calculators are **NOT** allowed in the examination.
- Periodic table is attached on the last page of the midterm examination.

Cheating in the exam is considered an extremely serious offence which will result in expulsion from the University.

	Multiple Choice	Part 2.1	Part 2.2	Total
score				

Part 1. Multiple Choice (10 points)

Choose the letter of the choice in the answer table that best completes the statement or answers the question.

Question no.	a.	b.	c.	d.
1				
2				
3				
4				
5				
6				
7				
8				
9				
10				

- Which of the following is NOT an example of matter?
 - air
 - smoke
 - heat
 - sea water
- A compound differs from a mixture in that the compound:
 - has two or more identities
 - has only one identity
 - is composed of only one kind of atom
 - can be separated by physical means.
- What is NOT true about evaporation?
 - Suitable for solids and liquids mixtures
 - Can be used to get salt from salty water
 - A matter having higher boiling point will be evaporated first
 - A matter having higher vapor pressure will be evaporated first
- A substance that can be separated into two or more substances only by a chemical change is a(n)
 - solution
 - mixture
 - element
 - compound
- According to the Aufbau principle, _____.
 - an orbital may be occupied by only two electrons
 - electrons in the same orbital must have opposite spins

- c. electrons enter orbitals of lowest energy first
- d. electrons enter orbitals of highest energy first
6. Which of the following situation is a chemical change?
 - a. A dish is broken.
 - b. Snow is melting.
 - c. A piece of paper is teared.
 - d. Rust of nails is growing.
7. The formula for magnesium sulfate is.....
 - a. MnS
 - b. MgS
 - c. $MgSO_4$
 - d. $MgSO_3$
8. Which of following statements is incorrect?
 - a. Sublimation is related to a heat releasing process.
 - b. Additional heat is not required for condensation.
 - c. Certain energy must be transferred to the substance via heat in freezing process.
 - d. Deposition can take place with or without heat.
9. An atom of phosphorus has an atomic number of 15 and a mass number of 31. How many neutrons does it contain?
 - a. 15
 - b. 16
 - c. 31
 - d. none of them.
10. Which of the following elements has the same number of unpaired electrons as sulfur (S)?
 - a. Oxygen (O)
 - b. Barium (B)
 - c. Phosphorous (P)
 - d. Oxygen (O) and phosphorous (P)

Part 2.1 Read the construction carefully and complete the information required.

(30 points)

1. Characteristics of solids, liquids and gases: Complete the table below.

The information can be used twice. (3 points)

Tightly packed, no definite shape or volume, definite volume but no definite shape, hard to compress, easy to compress, do not flow, flow easily, diffuse to fill all available space.

Solids	Liquids	Gases

2. Identify the following as a **chemical** or a **physical** change. (3 points)

a. When sugar dissolves into water	
b. Gasoline burns in the presence of oxygen gas.	
c. hydrochloric acid evaporating to hydrochloride gas	
d. Iron rusts, producing a layer of oxide on the outside of the metal.	
e. Water can boil at 100 degrees Celsius.	
f. Vinegar reacts with baking soda to create a new gas	

3. Fill the information to the table below (3 points)

Change of State	Description	Exothermic or endothermic Process
Solid – Liquid		
	Freezing	
	Boiling	
Gas – Liquid		
	deposition	
Solid – Gas		

4. Give the formulae of the following compounds. (5 points)

Sodium hydride

Copper (II) sulfate

Strontium phosphate

Barium hydroxide

Tetraphosphorus decaoxide

Lithium hydrogencarbonate

Potassium nitrate

Magnesium carbonate

Manganese (IV) oxide

Sulfur trioxide

5. Name the following compounds. (5 points)

NaCl _____
 AuPO₄ _____
 CO₂ _____
 Fe₂O₃ _____
 F₂O _____
 NH₄NO₃ _____
 HgCl₂ _____
 Li₂SO₄ _____
 SO₃ _____
 NH₃ _____

6. Match the best answer and put the answer into the blank. (3 points)

A. Evaporation	B. Filtration	C. Chromatography	D. Fractional distillation
E. Distillation	F. Using a magnet	G. Centrifugation	H. Crystallization
I. precipitation	J. Electrolysis	K. Sublimation	
_____	i. The best way to get salt from salty water		
_____	ii. Separating iron filings from a mixture of iron and sulfur		
_____, _____	iii. What is the correct order for obtaining salt from a mixture of sand and salt? (Provide 2 possible methods)		
_____	iv. If you wanted to make pure drinking water from sea water, what process would you use?		
_____	v. Used to separate heterogeneous mixtures composed of solids and liquids		
_____	vi. Crude oil can be separated into several liquids that have different boiling points. What is the name of this process?		

7. Choose the best answer and put the answer into the blank. (the answer can only be used once) (3 points)

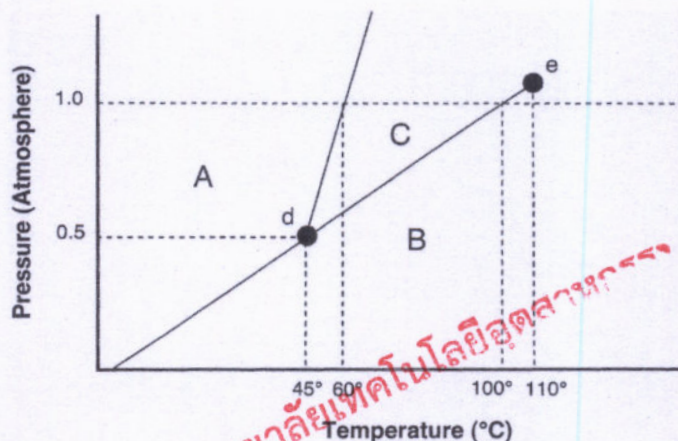
- | | |
|---------------------------|---|
| 7.1 Vinegar | a. The size of particle $> 10^{-4}$ cm |
| 7.2 Iron | b. The size of particle $< 10^{-7}$ cm |
| 7.3 CO ₂ | c. Their oxidation number is always +1 |
| 7.4 Si As | d. Chalcogens |
| 7.5 Dust | e. Tyndall effect |
| 7.6 O Se S | f. In the liquid state and have the electrical conductivity |
| 7.7 Hg | g. It comprises of 2 elements |
| 7.8 Li Na K | h. Weak conductivity |

..... 7.9 He Ne Ar

i. Their outer electron shell filled

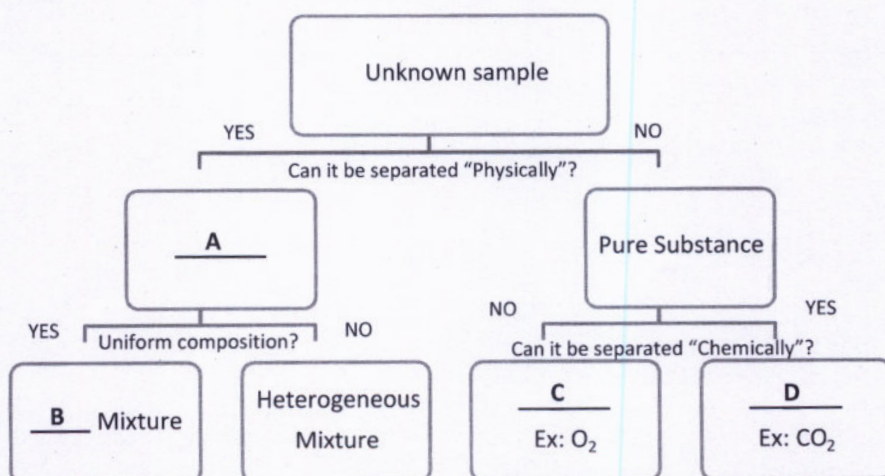
j. Have the electrical conductivity and the oxidation number is more than 1 values

8. Answer the questions below in relation to the following generic phase diagram. (3 points)



- Which section represents the gas phase? _____
- What letter represents the triple point? _____ In your own words, what is the definition of a triple point? _____
- What is this substance's normal melting point, at 1 atmosphere of pressure? _____
- At what temperature and pressure do all three phases coexist? _____
- At a constant temperature, what would you do to cause this substance to change from the liquid phase to the solid phase? _____

9. Provide the suitable words or phrases for A-D? (2 points)

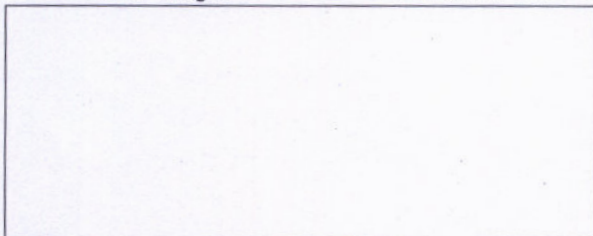


- A. _____ C. _____
- B. _____ D. _____

1.2 According to Rutherford's model, draw and specify all fundamental particles of fluorine atom (F) and fluoride ion (F^-). (4 points)

fluorine atom (F)	fluoride ion (F^-)

1.3 Draw a diagram for carbon (C) atom based on the Plum Pudding model (2 points)



2. Electron Configuration (18 points)

2.1 Write electron configuration of a chlorine atom and specify how many electrons occupy p orbitals? (2 points)

.....

2.2 Write electron configuration of a sulfur atom and specify how many electrons are related to the third energy level? (2 points)

.....

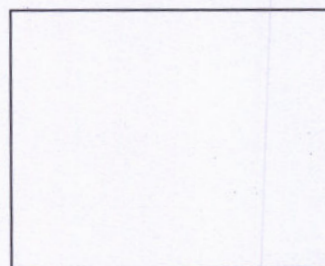
2.3 What elements have the following electron configurations? (2 points)

(a) $[Kr] 5s^2 4d^{10} 5p^1$

(b) $[Xe] 6s^2$

2.4 If principal quantum number (n) is equal to 1, what is possible angular momentum (l) and how does its orbital shape look? (2 points)

.....



2.5 Write all possible quantum numbers including n , l , m_l and m_s for Beryllium (Be).
(4 points)

.....

.....

.....

.....

2.6 Provide 2 characteristics of each following. (6 points)

(a) Elements in Group IA

.....

.....

(b) Noble gases

.....

.....

(c) Transition metals

.....

.....

Nathjanan Jongkon and Sunanta Chuayprakong

วิทยาลัยเทคโนโลยีอุตสาหกรรม

2019
IPT
International Year
of the Periodic Table
of Chemical Elements



United Nations
Educational, Scientific and
Cultural Organization



INTERNATIONAL UNION OF
PURE AND APPLIED CHEMISTRY

57	La	lanthanum	138.91
58	Ce	cerium	140.12
59	Pr	praseodymium	140.91
60	Nd	neodymium	144.24
61	Pm	promethium	
62	Sm	samarium	150.36(2)
63	Eu	europlum	151.96
64	Gd	gadolinium	157.25(3)
65	Tb	terbium	158.93
66	Dy	dysprosium	162.50
67	Ho	holmium	164.93
68	Er	erbium	167.26
69	Tm	thulium	168.93
70	Yb	ytterbium	173.05
71	Lu	lutetium	174.97
89	Ac	actinium	
90	Th	thorium	
91	Pa	protactinium	
92	U	uranium	
93	Np	neptunium	
94	Pu	plutonium	
95	Am	americium	
96	Cm	curium	
97	Bk	berkelium	
98	Cf	californium	
99	Es	einsteinium	
100	Fm	fermium	
101	Md	mendelevium	
102	No	nobelium	
103	Lr	lawrencium	